Notes

Minnetonka Math Team

Updated 16 August 2026

It is a maxim of math competitions that memorization matters less than problem-solving skill. However, there are some cases where memorization is important. Here are some things I should remember, but have trouble with.

1 Complex bashing

- Let the unit circle ω be tangent to $\overline{BC}$, $\overline{CA}$, and $\overline{AB}$ at D, E, and F, so ω is an incircle or excircle of $\triangle ABC$. Then the center of (ABC) is $\frac{2(d+e+f)def}{(d+e)(e+f)(f+d)}$, and the intersections between (DEF) and (ABC) are the roots of the two equivalent quadratics

$$2m_{11}x^2 - (6m_3 + m_{21})x + 2m_{211} = 0$$
$$\overline{a}(2d + a)x^2 - (d^2\overline{a} + 2d + a + 2da\overline{a})x + da(d\overline{a} + 2) = 0.$$

- Jacobi bialternant formula: $\frac{a_{\lambda+\rho}}{a_\rho} = s_\lambda$.

- Let k be a field of rational functions of some variables with coefficients in $\mathbb{Q}$ (for example, $\mathbb{Q}(a, b, c)$). Let $h \in k[x]$ have roots α and β. Then for a rational function $\frac{f}{g} \in k(x)$ with $\max(\deg f, \deg g) + 1 = n$, we have $\frac{f(\alpha)}{g(\alpha)} = \frac{f(\beta)}{g(\beta)}$ if the $n \times n$ determinant with one row as the coefficients of f, one row as the coefficients of g, and $n - 2$ rows as the coefficients of the coefficients of $x^i h(x)$ for each $0 \le i < n - 2$ is zero. In particular, if h is irreducible, this is an if and only if. The proof is by interpreting $k[x]$ as a vector space.

- In $\triangle ABC$, the A-Dumpty point D is the center of the pair of spiral similarities sending $\overline{AB}$ to $\overline{CA}$ and $\overline{AC}$ to $\overline{BA}$. In terms of complex numbers, $D = \frac{A^2-BC}{2A-B-C}$. In particular, if $A^2 = BC$, then $D = 0$. It may be checked (by setting (ABC) to be the unit circle) that $D \in (BOC)$, where O is the circumcenter of $\triangle ABC$.

- Let $ABCD$ be a convex quadrilateral inscribed in the unit circle. It is possible to choose complex numbers a, b, c, and d such that $A = a^2$, $B = b^2$, $C = c^2$, $D = d^2$, and the arc midpoints of $\widehat{AB}$, $\widehat{AC}$, $\widehat{CD}$, $\widehat{DA}$, $\widehat{ABC}$, and $\widehat{BCD}$ are $-ab$, $-bc$, $-cd$, $+da$, $+ac$, and $+bd$, respectively.

2 Other

- For an odd prime p and $0 \le n \le \frac{p-1}{2}$, we can transform $\binom{2n}{n} \equiv (-4)^n \binom{\frac{p-1}{2}}{n} \pmod{p}$.

- Factorials modulo p are almost always dealt with considering inverses and exploiting the symmetry $x \mapsto -x$. Consider $(\frac{p-1}{2})!$ often.

3 Estimation

We begin with a log table. Here log is the natural log.

log	value
$\log 2$	0.693147
$\log 3$	1.09861
$\log 5$	1.60944
$\log 7$	1.94591

You will usually only need the first three significant figures, but sometimes problems are weird.

We continue with some facts about space. I don't believe in significant figures.

- The speed of light is $3 \cdot 10^8$ meters per second.
- One AU is 8 light-minutes. It takes light 1.3 seconds to reach the moon.
- The radius of the Earth is 6000 kilometers.
- Mercury, Venus, Earth, Mars, the asteroid belt, Jupiter, Saturn, Uranus, Neptune, and Pluto are $\frac{1}{3}$, $\frac{2}{3}$, 1, 1.5, 3, 5, 10, 20, 30, and 40 AU from the Sun, respectively.
- The Sun has a mass of $2 \cdot 10^{30}$ kilograms.
- Jupiter weighs $\frac{1}{1000}$ as much as the Sun. Saturn weighs $\frac{1}{3}$ as much as Jupiter. Uranus and Neptune weigh $\frac{1}{6}$ as much as Saturn, or 10^{26} kilograms.
- Earth weighs $\frac{1}{1000}$ as much as Saturn. Venus weighs as much as Earth. Mars weighs $\frac{1}{10}$ as much as Earth. Mercury weighs $\frac{1}{2}$ as much as Mars. The moon weighs $\frac{1}{100}$ as much as Earth.
- The sun and the gas giants have a density of around 1.4 grams per cubic centimeter, except for Saturn, which is around half that. The rocky planets have a density of around 5 grams per cubic centimeter.
- The Milky Way has a diameter of 10^5 light years. Andromeda is 25 Milky Way diameters away. If we were to flatten the Milky Way into a uniform disk, we would have 50 solar masses per square light year.